SLAM-Assisted Transformer-Based TD3 Deep Reinforcement Learning for Adaptive Navigation and Mapping and Dynamic Obstacle Avoidance in Autonomous Construction Robotics

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ABSTRACT

This research introduces a SLAM (simultaneous localization and mapping)-assisted deep reinforcement learning (DRL) framework that integrates a transformer-based twin delayed deep deterministic policy gradient (TD3) for efficient navigation and obstacle avoidance in differential drive robots. A real-time SLAM is integrated to provide accurate localization, real-time occupancy grids, and precise robot pose estimates, which are integrated into the TD3 reward estimation to enhance navigation precision. Key advancements include the integration of transformer-based architecture within TD3 to enhance generalization and robustness in dynamic construction environments. The system is trained in a simulated environment using ROS2 and Gazebo with domain randomization to introduce variations in lighting, sensor noise, and dynamics, thus bridging the simulation-to-reality gap (Sim2Real). Key findings include a 90% task success rate in navigating complex construction site layouts with dynamic obstacles. The enhanced TD3 model demonstrated significant adaptability by navigating unpredictable layouts and avoiding continuously changing obstacles while maintaining operational efficiency.

INTRODUCTION

Automation and digitalization in the construction industry have recently surged, especially with the increasing complexity of modern construction projects in the Architectural, Engineering, and Construction (AEC) industry (Ametepey et al. 2024). As global infrastructure needs continue to rise, the demand for safer, more efficient and less labor-intensive methods of construction has grown correspondingly. Construction 5.0 envisions a human-centered automated and intelligent construction environment which positions robotics and advanced sensing technologies at the forefront of transformational change in civil engineering operations (Musarat et al. 2023). The integration of robotics and artificial intelligence (AI) into the construction sector aligns with the pressing industry imperatives including improved worker safety, enhanced productivity, and reduced operational costs. Autonomous construction robots generally navigate and operate in dynamic, unstructured and hazardous environments. Constantly shifting layouts, moving equipment, variable lighting and unpredictable obstacles complicate navigation tasks (Trevelyan et al. 2016). Static path planning techniques and conventional navigation algorithms frequently struggle to cope with these complexities resulting in performance degradation and compromised safety. Recent advancements in Reinforcement Learning (RL) particularly Deep Reinforcement Learning (DRL) present a promising approach to address the challenges (Wang et al. 2022). RL, as illustrated in Figure 1, is a type of machine learning where an agent interacts with an environment through an interpreter to process the state information.

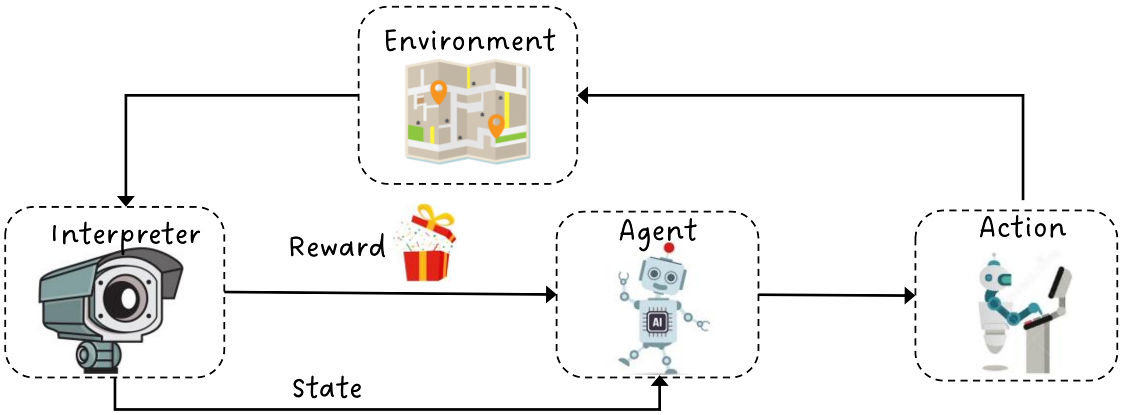


Figure 1. Reinforcement learning general framework

The agent takes actions within the environment to achieve a goal, receiving feedback in the form of rewards, and iteratively learns to maximize cumulative rewards by refining its decision-making policy (Ding et al. 2020). DRL combines RL with deep learning enabling agents to make decisions from high-dimensional sensory inputs, such as lidar without manual feature engineering. DRL enables autonomous agents to learn optimal navigation policies through continuous interaction with their environment rather than relying on predefined rules. Techniques like the Twin Delayed Deep Deterministic Policy Gradient (TD3) have demonstrated robust learning capabilities and improved sample efficiency compared to earlier DRL algorithms (Fujimoto et al. 2018). However, standard DRL models often grapple with limited generalization and adaptability when confronted with significant environmental uncertainties and domain shifts common to real-world construction scenarios.

On the other hand, simultaneous Localization and Mapping (SLAM) has emerged as a critical technology to enhance the situational awareness of autonomous robots (Bresson et al. 2017). SLAM algorithms provide accurate localization estimates and up-to-date occupancy maps of the environment with accurate robot's state representation. Integrating SLAM with DRL stabilizes policy learning and allows the agent to understand spatial constraints more effectively for safer and more efficient navigation in complex settings. In parallel, the use of transformer-based architectures into DRL frameworks holds substantial promise as it excels in capturing long-range dependencies and contextual information (Jezzini et al. 2025). Their adaptability and robustness have recently found success in various AI domains, and their use in DRL scenarios is gaining traction (Hu et al. 2024). This architecture can help autonomous agents better generalize to unseen conditions, making them particularly suitable for environments as unpredictable as construction sites.

This research introduces a novel SLAM-assisted, transformer-based TD3 DRL framework for autonomous navigation and obstacle avoidance in dynamic construction environments. By leveraging SLAM for accurate localization and mapping and integrating transformer-based models for improved policy robustness, this approach aims to bridge the gap between simulation and reality. The proposed system demonstrates superior adaptability, safety, and efficiency for successful deployment in the construction industry. This work advances the state of the art in autonomous construction robotics, paving the way for safer, more reliable, and intelligent solutions in civil engineering operations.

GOAL AND OBJECTIVES

The goal of this paper is to develop a robust, SLAM-assisted, transformer-based reinforcement learning framework that enables autonomous construction robots to efficiently navigate, adapt, and operate in dynamically evolving construction environments. The research objectives include: (i) Integrating a real-time SLAM system with a Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm to enhance the robot's spatial awareness and improve navigation accuracy within dynamic construction site conditions; (ii) Incorporating transformer-based architecture into the TD3 framework to strengthen policy generalization, enabling the robot to maintain performance levels despite varying lighting conditions, sensor noise, and unpredictable obstacle patterns; and (iii) validating the developed framework in simulation environments with domain randomization.

LITERATURE REVIEW

Recent trends in the construction industry show the increase of robotics and autonomous systems to address the escalating challenges in workforce shortages, safety concerns, and the need for higher productivity (Bademosi et al. 2022). As construction sites are becoming more complex and dynamic (Poudel and Assaad, 2024), researchers have explored various approaches to ensure reliable robotic navigation, obstacle avoidance, and adaptability to uncertain conditions. Traditional navigation algorithms like A* and D* provide efficient solutions in well-structured environments like in factory settings (Gul et al. 2019). These classical methods generally rely on predefined maps and struggle in construction scenarios where the environment is dynamic and cluttered with equipment, materials and transient obstacles (Hoy et al. 2015; Melenbrink et al. 2020). Moreover, approaches relying on static maps frequently underperform when confronted with sudden layout changes, varying lighting conditions, or inclement weather, all of which are commonplace on construction sites (Agostinho et al. 2022). To overcome these limitations, reinforcement learning comes out as a strong alternative due to its data driven capability to learn navigational policies through trial and error. Studies like Zhang et al. (2018) and Liu et al. (2020) are among the first works exploring the use of DRL for mobile robot navigation. The TD3 algorithm emerged as a significant advancement in off-policy actor-critic DRL methods which addressed several limitations found in earlier frameworks like DDPG (Shen 2024). By using twin Q-networks to mitigate overestimation bias and employing delayed policy updates, TD3 achieves greater stability, robustness, and sample efficiency in continuous control tasks (Fujimoto et al. 2018). Although TD3 has shown promise in simulation-based robotics problems, transitioning to real-world environments especially those as variable and unpredictable as construction sites remains challenging. To strengthen the generalization capabilities and adaptability of TD3, researchers have begun to integrate more sophisticated neural network architectures into its design (Lu et al. 2025). In parallel, robotic perception has advanced through the development of Simultaneous Localization and Mapping (SLAM) algorithms, which allow autonomous agents to build and maintain a coherent representation of their environment while estimating their own position (Rosen et al. 2021. Integrating SLAM outputs like occupancy grids into RL-based navigation strategies has shown potential in enhancing state representations and stabilizing learned policies (Mustafa 2019). This integration helps the robot's decisions; not only guided by immediate sensor readings but also by a continuously updated global map to improve long-term navigation performance and adaptability. While SLAM and DRL offer promising avenues, one of the key challenges remains the generalization of learned policies across varying environments and conditions. Transformer-based architecture has recently emerged as a potent tool in this regard. Huang et al. (2023) demonstrated that transformers could enhance policy robustness and generalization, enabling agents to handle more complex state and action spaces than conventional convolutional or recurrent networks. By capturing global context and relationships, transformers appear well-suited to dynamic and uncertain environments, offering a potential breakthrough in DRL-based navigation tasks within construction settings (Hu et al. 2024). Another critical aspect of real-world deployment is the simulation-to-reality (Sim2Real) gap. Domain randomization techniques in which simulation parameters such as lighting, obstacles, or noise levels are altered across training episodes have shown improvement over the transferability of learned policies to real-world conditions (Shakerimov et al. 2023).

The existing literature has laid vital groundwork in the domains of autonomous navigation, SLAM integration, and TD3 based control. However, a gap remains: few studies have jointly leveraged SLAM, transformer-enhanced TD3 frameworks, and domain randomization techniques to create an integrated, robust navigation and mapping system tailored specifically for the complexities of construction environments. This research aims to fill that gap by proposing and validating a SLAM-assisted, transformer-based TD3 reinforcement learning framework designed for autonomous construction robotics.

METHODOLOGY

This section describes the overall experimental setup, from the simulation environment and mapping system to the integration of a transformer-enhanced TD3 framework and its evaluation.

Simulation Environment and Domain Randomization: Simulation environment is set up using Gazebo robot simulator with ROS2 as backend as shown in Figure 2.

The robot platform used is a TurtleBot 3 Waffle PI; a widely used differential-drive mobile robot equipped with a 2D LiDAR sensor and onboard computation resource. Various domain randomization techniques were used to ensure that the learned policies generalize well and do not overfit. The sensor reliability was also perturbed; with LiDAR readings being subjected to randomized noise representing air with different particle densities, or partial occlusions because of dust or structures. Furthermore, the introduction of dynamic obstacles or mobile entities that move unpredictably through the workspace simulates the presence of human workers or shifting equipment common on a construction site. This require the robot to adapt in real time instead of relying on static assumptions. The training process, by constantly varying the factors involved from obstacle arrangements and sensor fidelity and mobile obstacles, results in a policy that can gracefully handle the intrinsically uncertain and fluid nature of real-world construction environments.

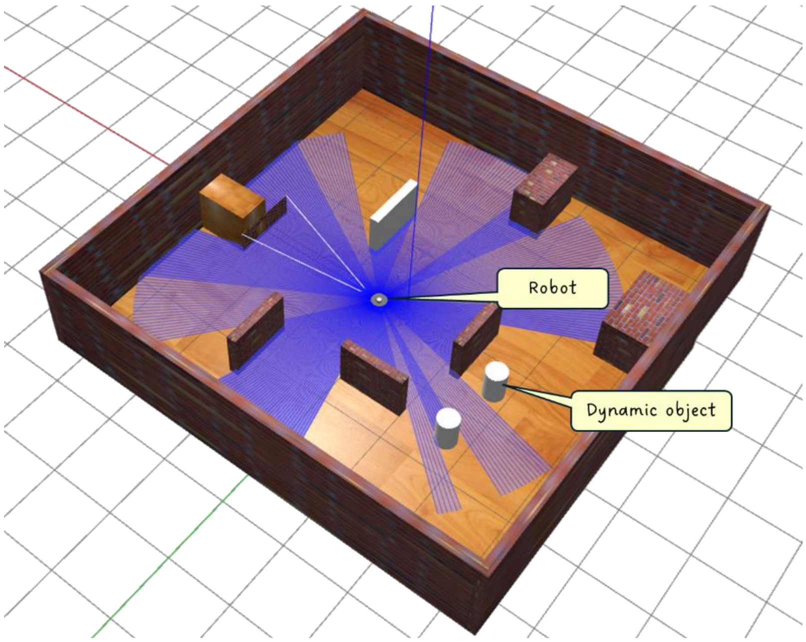


Figure 2. Developed simulation environment in Gazebo

SLAM Integration and Mapping: Accurate mapping and localization are essential for robust and adaptive navigation strategies in complex environments. For the SLAM module, Google Cartographer SLAM algorithm was used. By continuously processing incoming lidar measurement $\left(z_{(1:t)}\right)$ and know control inputs $\left(u_{(1:t-1)}\right)$, SLAM incrementally estimates the robot's pose x_t and maintains a probabilistic occupancy grid map M_t of the environment. The SLAM problem can be expressed as inferring the posterior distribution over the robot's current pose given all past observations and actions as shown in Equation 1.

$$\mathrm{p}(x_t \mid z_{1:t}, u_{1:t-1}) = \int p(x_t \mid x_{t-1}, u_{t-1}) p(x_t - 1 \mid z_{1:t-1}, u_{1:t-1}) dx_{t-1} \quad (1)$$

The algorithm solves this recursive estimation problem by an occupancy grid map Mt: $X \subseteq R^2 \rightarrow [0,1]$, where each cell in the grid represents the probability of that cell being occupied by an obstacle. As the robot explores and collects more sensor data, M_t is continually refined to enhance global spatial awareness and allowing the robot to make more informed navigation decisions. The constructed map; M_t actively updates the RL algorithm through map driven reward shaping mechanism. At each timestep t, the robot observes state S_t, take an action a_t, and received a reward r_t that encourages effective navigation strategies. This reward function is designed to incorporate environmental understanding derived from M_t.

Equation 2 gives $\mathrm{C}(M_t)$; a converge metric to quantity the fraction of the designated region of interest (ROI) that has been successfully mapped.

$$\mathrm{C}(Mt) = \frac{count(all\ cells\ in\ the\ ROI)}{count(mapped\ cells\ in\ Mt)} \quad (2)$$

The reward is structured as shown in Equation 3.

$$\mathrm{r_t} = r_{base}(s_t, a_t) + \lambda_{map} \cdot f_{map}(M_t) \quad (3)$$

Where, $r_{base}(s_t, a_t)$ provides fundamental navigation incentives (reward metric in RL algorithm except the mapping like obstacle avoidance, moving closer to goal location etc.). The term $f_{map}(M_t)$ (incremental coverage gain) leverages the converge metric $\mathrm{C}(Mt)$ to reward exploration into previously unmapped or uncertain territories improving the robot's long-term autonomy and situational awareness. λ_{map} is a weighting factor to scale the mapping-based reward metrics. By modeling an evolving occupancy map-based reward function, the learning process becomes effectively coupled with the robot's growing understanding of its environment. This approach ensures that the agent learns not only to move efficiently but also seeks out new areas to map, refining its spatial knowledge, and avoiding risks identified in the environment model.

Transformers based Reinforcement Learning Algorithm: Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm was used for navigational tasks, as illustrated in Figure 3. The algorithm is known for its stability in continuous control tasks. In TD3, two critic networks are maintained: $Q_{\theta 1}$ and $Q_{\theta 2}$ to reduce overestimation bias and an actor network π_ϕ that outputs continuous actions. The target for the critical update is shown in Equation 4.

$$\mathrm{y_t} = r_\mathrm{t} + \gamma \min_{j=1,2} Q_{(\theta' j)}\left(s_{(t+1)}, \pi_{(\emptyset)}\left(s_{(t+1)}\right) + \epsilon\right) \tag{4}$$

Where γ is the discounting factor, θ'_j and $\emptyset$ denote parameters of the target networks which are softly updated from the main networks and $\in$ is clipped noise to improve policy smoothness. The critic loss is given in Equation 5.

$$L_Q(\theta) = E[(Q_{\theta 1}(s_t, a_t) - y_t)^2 + (Q_{\theta 2}(s_t, a_t) - y_t)^2] \tag{5}$$

The actor is updated by maximizing the Q-value under the first critic as shown in Equation 6.

$$L_\pi(\phi) = -E\left[Q_{\theta 1}\left(s_t, \pi_\phi(s_t)\right)\right] \tag{6}$$

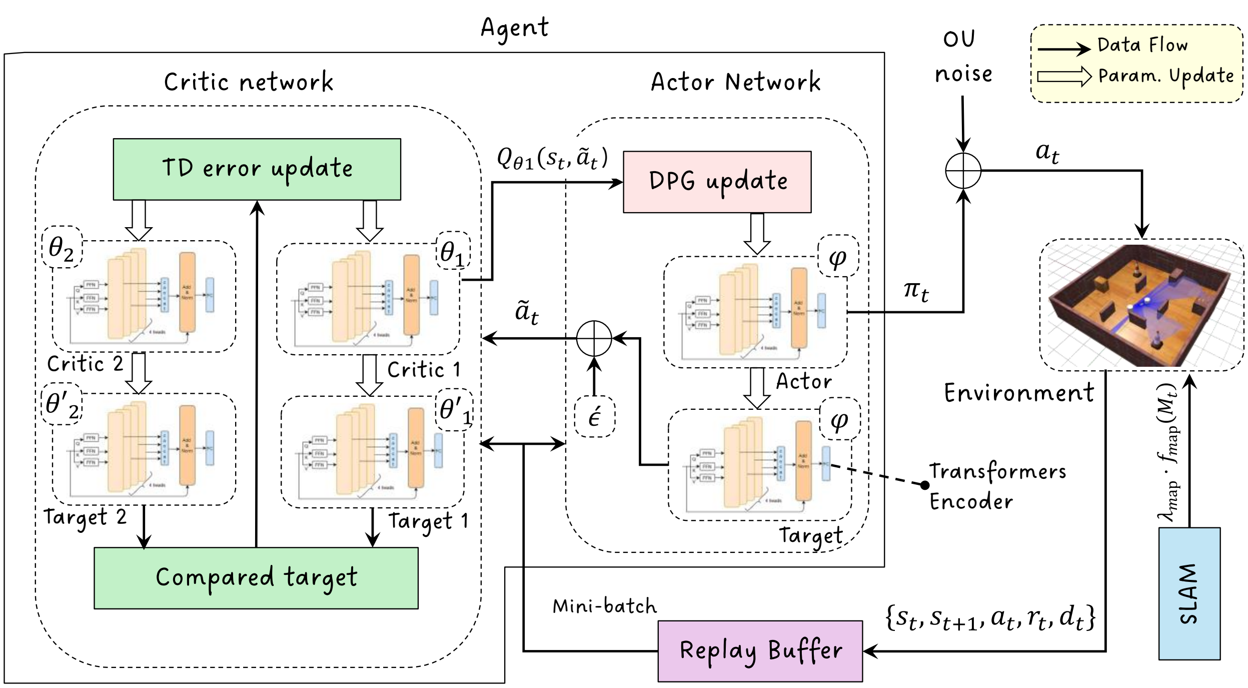


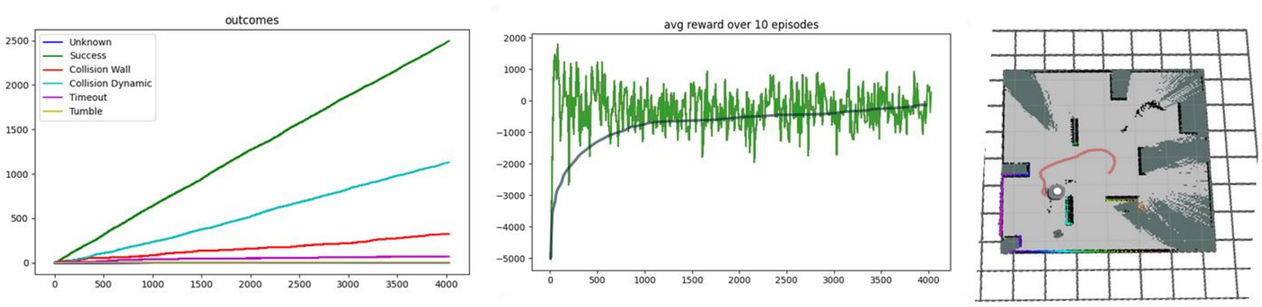
Figure 3. Architecture of the developed SLAM assisted Transformers based TD3 algorithm

To enhance TD3's capacity to handle complex layout, a transformer encoder was integrated into the actor and critic network. The sensor information is first linearly projected into a fixed embedding space before passing through a series of encoder layers. The layers consist of positional encodings to inject the notion of order ensuring time step are meaningfully distinguishable. Multi-headed self-attention module in encoder enables the network to learn context rich representation focusing on different parts of the input sequence. By modeling global dependences, the attention mechanism strengthens the network to relate specific sensor measurement with previously mapped area or previous readings. The resultant representation is aggregated to create a context aware feature vector. In the actor network, this vector informs the action selection process and in critic networks, the representation enables more accurate Q-value estimation. Transformers enhance TD3 by discerning essential patterns and dependencies from higher dimensional sequential inputs making network more flexible and robust for navigation in dynamic environments.

Training and Evaluation Metrices: Each episode in training starts by placing the robot in the origin and randomly selecting a goal position within the defined area boundaries. The agent (robot) interacts with the environment at each timestep by observing the current state, selecting an action (linear and angular velocities), executing the action and receiving a reward signal. During the beginning of the training for *"n"* defined steps, the agent acts randomly to encourage wide range exploration and populate the reply buffer with diverse set of experience tuples (s, a, r, s', d). After this the agent samples mini batches from the reply buffer to update the actor and critic networks using TD3 algorithm. The system continuously evaluates and logs performance metrics. Key training parameters include learning rate set at 0.004, discounting factor of 0.99, replay buffer size of 1,000,000 samples, action noise and policy noise clipping of 0.2 and 0.5 respectively, and timeout of 60 seconds. Multiple metrics were used to evaluate the performance of the RL algorithm. Success rate quantifies how often the robot reaches its goal within a specified threshold, showing the task completion capability. Collision frequency is a safety measure in terms of counting obstacle impacts per episode. Finally, monitoring the trends in rewards shows the policy's generalization capability. These collectively provide a complete view of the effectiveness, safety, and resilience of the policy within complex dynamic construction scenarios.

RESULTS AND ANALYSIS

The proposed SLAM-assisted, transformer-based TD3 framework was evaluated in a simulated construction environment after training for 4,000 episodes. Throughout training, the agent interacted with the environment, accumulating experience and refining its policy using the integrated SLAM maps and transformer-based representations. During the training phase, a trend toward improved performance emerged. Figure 4(a) displays cumulative outcomes over the course of training episodes showing that the agent gradually increased its success rate in reaching designated goals. Early in training, the robot frequently encountered collisions and other failure modes due to the complexity and variability of the site conditions. However, as the policy matured, the incidence of undesired events (wall collisions, obstacle impacts, and timeouts) declined, and successful completions steadily rose. By the end of the training period, success rates approached the target performance levels set out in the objectives.



(a) (b) (c)

Figure 4. (a) Training outcomes after 4,000 episodes, (b) Reward accumulation after 4,000 episodes, (c) SLAM sample visualized in Rviz

Concurrently, the average reward illustrated in Figure 4(b) stabilized into a narrower band as training progressed. Initially volatile due to extensive exploration and the complexity of the domain-randomized environment, the reward curve's convergence indicates that the agent learned consistent strategies for safe and efficient navigation. Qualitative analysis of the generated occupancy maps (Figure 4(c)) also supports these findings. Over time, the robot constructed more coherent and detailed environmental representations, leveraging these maps to identify safer, more direct routes. The advanced transformer-based actor and critic networks capitalized on these improved representations to guide the agent toward stable, high-performance policies. To further validate the learned policy, a dedicated testing phase was conducted over 1,000 evaluation episodes and compared with the standard neural network based TD3 from Cimurs (2022) tested in the same environment. In these tests, the agent maintained a 90% success rate in reaching the goal, while only 3% of episodes resulted in wall collisions, 1% ended due to timeout, and 6% involved collisions with dynamic obstacles where pretrained standard neural network based TD3 got 82% success rate, 6% wall-collision, and 3% timeout and 9% collision with dynamic obstacles. The post-training performance demonstrates the generalization capability and robustness of the proposed method in comparison with standard TD3. The agent successfully transferred its learned skills to unseen scenarios, confirming that the integrated SLAM and transformer-based approach substantially improves navigation accuracy, reliability, and adaptability. In summary, the results show that the SLAM-assisted, transformer-based TD3 framework met the primary research objectives. It provides robust and efficient navigation solutions under domain-randomized conditions, achieving high success rates, reduced collision frequencies, stable reward patterns, and improved path planning efficiency.

DISCUSSIONS

The results indicate that integrating transformer-based architecture into a TD3 framework enhanced by SLAM-based mapping and a map-driven reward structure leads to noticeable improvements in autonomous navigation within simulated construction environments. Unlike a conventional TD3 approach relying on standard feedforward neural networks, the transformer-enhanced method better leverages sequential sensor information, continually updated occupancy maps, and positional encodings to identify critical patterns in real-time. As a result, the agent navigates more efficiently, avoids obstacles more reliably, and adapts more readily to domain-randomized conditions that mimic the variability and complexity of actual construction sites. A key advantage of this approach lies in its ability to interpret LiDAR scans, and map-based features as contextual sequences rather than isolated events. Transformer encoders employ self-attention mechanisms to relate every part of the input to every other part and thus the network highlight important elements (such as subtle changes in obstacle distribution) while disregarding irrelevant noise. In contrast, a standard TD3 network would struggle to capture these relationships as effectively. By fusing SLAM outputs with transformer-driven policy and value networks, the agent establishes a strong spatial and temporal awareness that translates into more direct paths to goals, fewer collisions, and a higher overall success rate in reaching targets within a given time frame.

The implications of these findings are substantial. Construction sites are inherently dynamic, with shifting layouts, temporary barriers, and unexpected human activity. A navigation policy that can generalize to such conditions and maintain stable performance helps pave the way for safer and autonomous operations in the real world. The combination of SLAM-based mapping and transformer architecture addresses the need for robots that do not simply follow predefined routes but rather understand and adapt to the environment as it evolves. This adaptability can reduce downtime, mitigate risks, and enhance efficiency for tasks such as material delivery, inspection, and data collection. From a theoretical perspective, this work contributes to the growing body of research exploring how advanced architectural choices, like transformers, can improve RL outcomes in continuous control tasks. It shows that going beyond conventional feed-forward networks allows for a richer understanding of the environment and encourages more robust policy formation. Although certain limitations remain such as the gap between simulations and the full complexity of real-world conditions, and the computational demands of transformer layers, the benefits demonstrated here argue strongly for further exploration and refinement. The findings highlight the promise of attention-based models in robotics and point toward practical advantages when deploying autonomous systems to navigate dynamic, real-world environments.

CONCLUSIONS

This paper addressed the challenge of achieving robust, efficient, and adaptive autonomous navigation in dynamic construction environments. Such settings, which are characterized by shifting layouts, unpredictable obstacles, and variable sensory inputs, pose a significant challenge for traditional robotic navigation methods. The research tackled the problem of safe and efficient robot navigation while continuously adapting to the uncertain, evolving conditions found on real construction sites. The Transformer encoder was integrated into a Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm and the strengths of SLAM-based mapping were combined with the reward structure. Unlike conventional feedforward networks, transformers capture long-range dependencies and contextual relationships within sequential data. By integrating self-attention mechanisms from transformers, the system could dynamically focus on the most pertinent elements of the robot’s sensory inputs and environmental maps to adjust its navigation decisions in response to subtle changes. The results demonstrated that the enhanced approach outperformed a standard neural network-based TD3 method in terms of success rates and collision avoidance under domain-randomized conditions that simulated the complexity and uncertainty of real construction scenarios. The findings suggest that practitioners can improve safety, reduce downtime, and increase overall productivity by adopting such approaches in actual construction operations. For the broader research community, these results underscore the potential of exploring attention-based models, more complex sensory integrations, and larger-scale deployments. By continuing to refine these methods, future work can bring the field closer to seamlessly deploying autonomous robots in real-world environments where adaptability, safety, and efficiency are paramount.

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